SYSTEM ARCHITECTURE · END-TO-END

Nabz نبض

A voice-first, Urdu-first AI health-triage companion. The browser captures speech; a React SPA talks only to a FastAPI backend; the backend owns every AI call, safety guardrail, and byte of patient data. It triages urgency — it never diagnoses or prescribes.

Frontend React 18 · Vite · RTL

Backend FastAPI · Pydantic v2

AI Qwen + Qwen-VL · DashScope

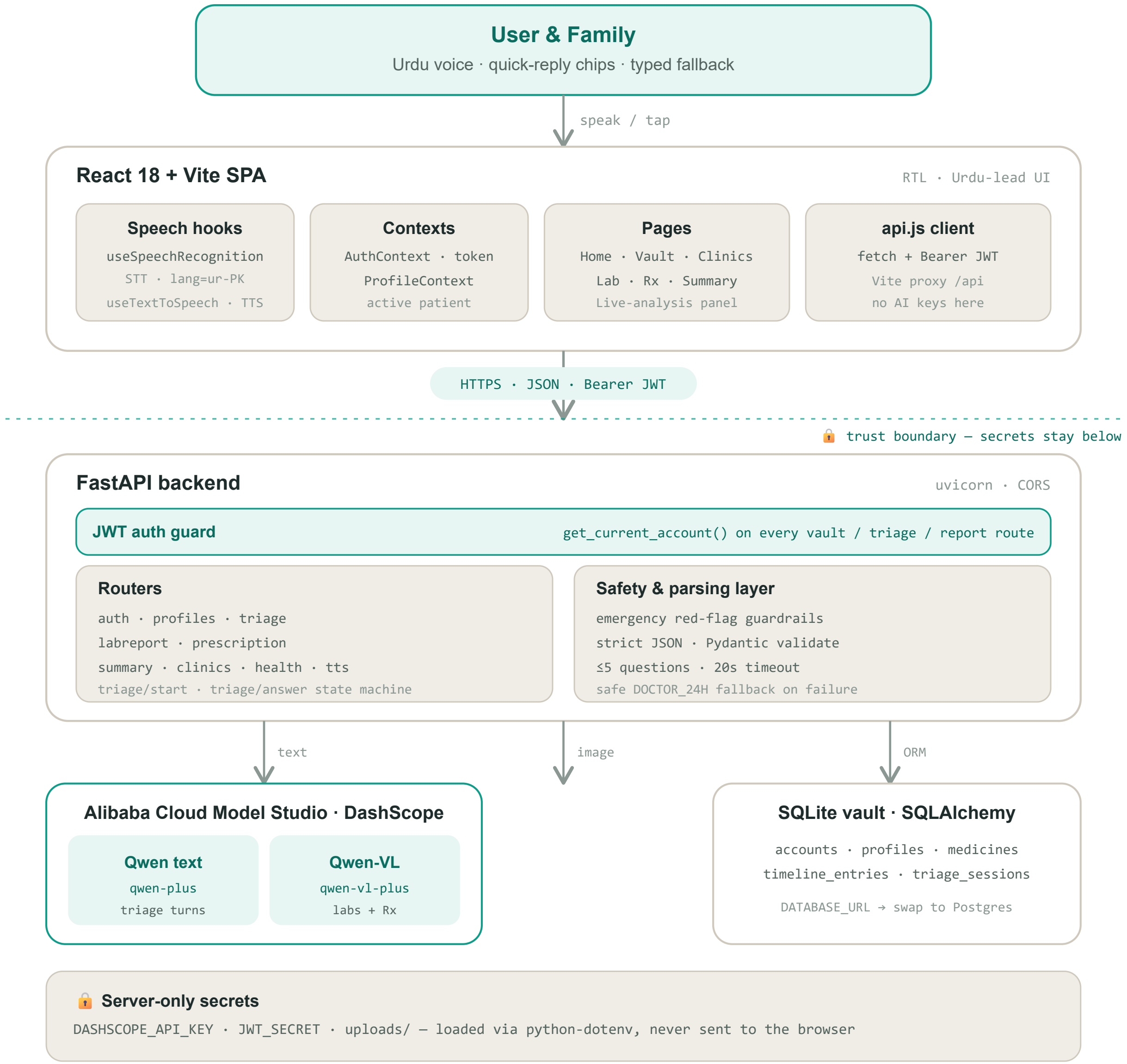
Data SQLite · SQLAlchemy

Auth JWT · bcrypt

01

The request path, top to bottom

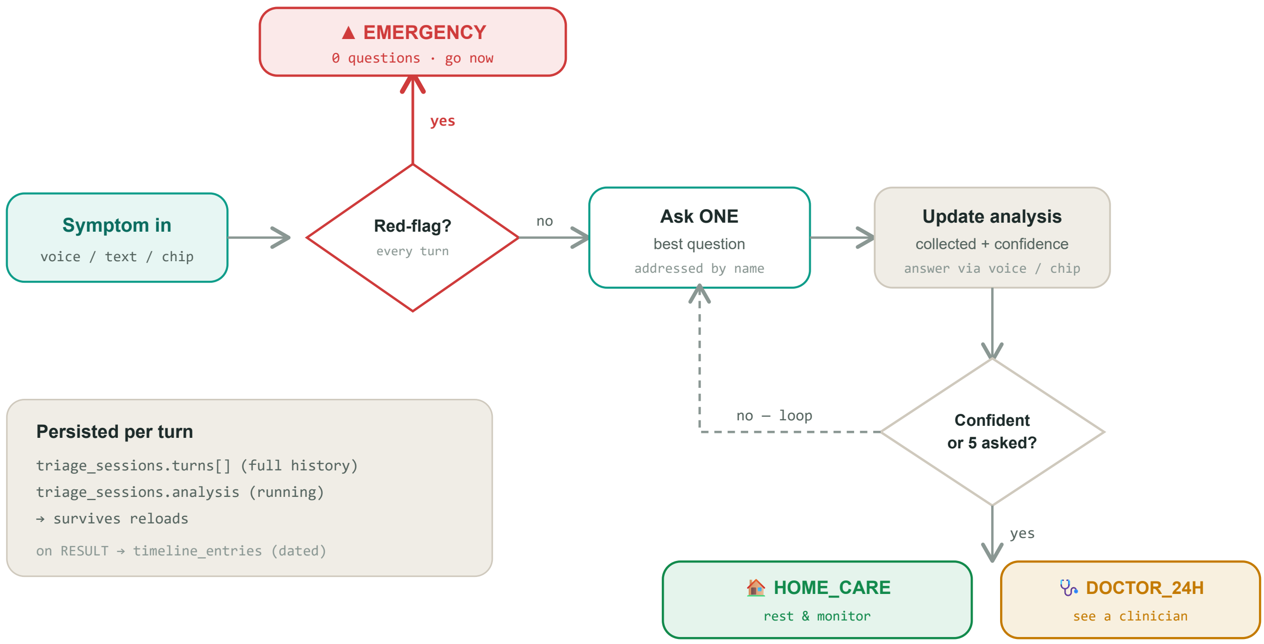
Every user action flows down one spine: voice or tap in the browser, through the React SPA, across a single JWT-authenticated JSON boundary, into FastAPI — which fans out to the Qwen models and the SQLite vault. The AI keys live only below that boundary.



One JSON boundary. The SPA holds a JWT but no model credentials; every Qwen call and the entire patient vault sit behind FastAPI. Remove the dashed line and the security story is the whole point — the browser can never reach DashScope directly.

02 Conversational triage — the state machine

Nabz does not jump from a symptom to a verdict. It asks the single highest-value question at a time, builds a live analysis, and only concludes when confident — **unless** a red flag appears, which short-circuits straight to EMERGENCY with zero further questions.



The red-flag check runs on every turn, not just the first — breathlessness reported on question three still escalates instantly. Uncertainty after five questions resolves upward to at least DOCTOR_24H, never down. The active profile's name, conditions and medicines are injected into each turn.

03 Data model — the family vault

One account owns many patient profiles; everything else hangs off a profile and is deleted with it (cascade). Personalization always pulls from the active profile **alone** — never a shared pool.

`accounts`

`1 ──▶ ∞ profiles`

full_name, **phone** (unique)

password_hash · bcrypt

created_at

`profiles`

`▶ medicines · timeline · sessions`

display_name, relation, age, gender

blood_group, is_self

chronic_conditions[], allergies[]

notes

`medicines`

`∞ ◀── profile`

name, strength, frequency

duration, with_food, notes

source manual | prescription

`timeline_entries`

`∞ ◀── profile`

kind triage | lab | prescription

title, subtitle, level

payload JSON

`triage_sessions`

`∞ ◀── profile`

turns[], analysis JSON

status, result_level

initial_text, result_payload

`uploads/`

`filesystem, not DB`

original lab & Rx images

referenced by timeline payload

view the source paper later

04 Mock mode vs. real mode

The whole app is demoable with zero credentials. A single gate — `is_mock_mode()` = `MOCK_MODE` truthy **or** no API key — decides which backend answers. Every AI response carries `"mock": true|false`.

Mock backend

`when MOCK_MODE=true OR key is missing`

- Deterministic keyword-based triage heuristic
- Canned follow-up questions & live analysis
- Sample CBC lab (flagged Hb / Iron)
- 2-medicine Rx extraction, one low-confidence field
- No network, no credentials — ideal for demos & CI

Real backend

`when DASHSCOPE_API_KEY set & MOCK_MODE=false`

- qwen-plus for conversational triage turns
- qwen-vl-plus for lab reports & prescriptions
- OpenAI-compatible SDK → DashScope intl endpoint
- Same guardrails still enforced outside the model
- Bad / untrusted output → safe DOCTOR_24H fallback

05 API surface

All routes except auth, clinics and health require a bearer JWT.

METHOD	ENDPOINT	PURPOSE	GUARD
POST	/api/auth/register · /login	Create account (+ self profile) · issue JWT	public
GET	/api/auth/me	Current account	jwt
GET·POST	/api/profiles	List / create family profiles	jwt
···	/api/profiles/{id}	GET · PUT · DELETE (+ all its data)	jwt · owner
POST	/api/triage/start · /answer	Conversational triage state machine	jwt
POST	/api/labreport	Qwen-VL lab extraction + Urdu explain	jwt
POST	/api/prescription · /confirm	Extract → user reviews → save to profile	jwt
GET	/api/summary/{id}	English doctor-handoff summary	jwt
GET	/api/clinics · /health · /tts	8 KP facilities · status · spoken Urdu	public

Safety boundary. Nabz triages urgency and reads documents — it never states a diagnosis, never prescribes, and never delays an emergency to keep asking. Prescription scanning only transcribes an existing clinician's paper and requires human confirmation before saving.